

राष्ट्रीय इलेक्ट्रॉनिकी एवं सूचना प्रौद्योगिकी संस्थान (रा.इ.सू.प्रौ.सं.), चेन्नई

National Institute of Electronics and Information Technology, Chennai

Autonomous Scientific Society of Ministry of Electronics & Information Technology (MeitY), Govt. of India

ISTE Complex, 25, Gandhi Mandapam Road, Chennai - 600025

Course Prospectus

“Cyber Security Associate” (NSQF - Level 4.5)

Mode: Online Blended



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Course Prospectus

Course Name: Cyber Security Associate

Course Code: IS181

NSDA CODE: 2021/ITES/NIELIT/00352

NSQF LEVEL: 4.5

Duration: 540 Hrs. (4 Months)

Last Date of Registration: 08th September 2025

Date of publishing Provisional Selection List: 09th September 2025

Payment of first instalment fee: 09th September 2025 to 10th September 2025

Course Start Date: 10th September 2025

Preamble:

Business @ the speed of thought has moved from a buzzword to a business reality owing to the integration of business and technology. Digital infrastructures and information networks have become central to any business activity. The information residing on these computers, networks and in cloud are an essential asset and due care should be taken for its security. The impact of any loss of this asset or any downtime of the infrastructure is quite high with the proliferation of easy-to-use tools and open how to- discussions, the number of people aiming to hack in to or bring down the network is high, both in this country and abroad. Hence, there is a need for heightened security measures to protect both infrastructure and data.

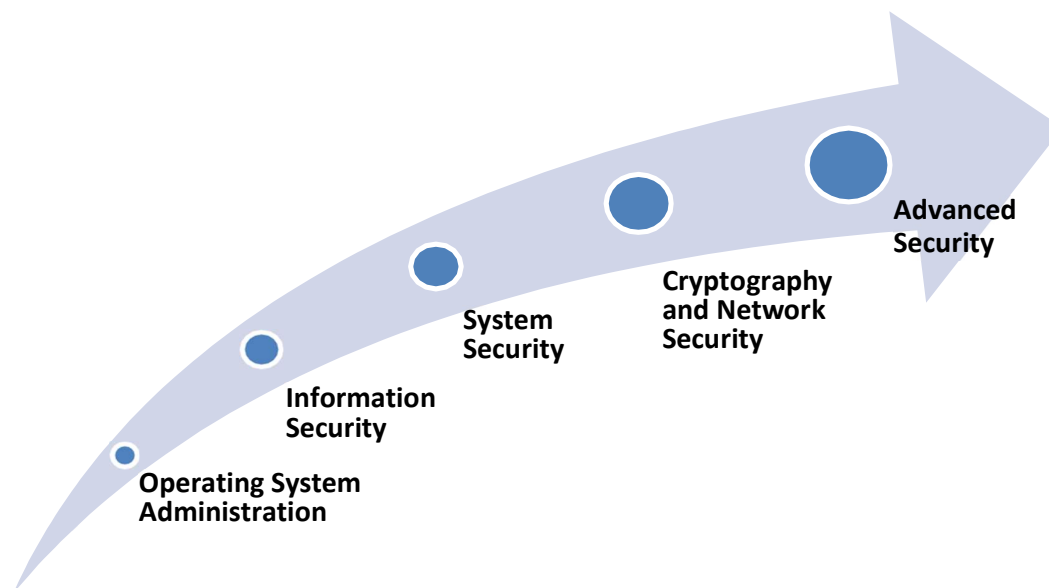
Objective of the Course:

The Course is 540 Hrs. (14 weeks) programs aimed at molding candidates to be a skilled knowledgeable System Security to maintain the Security of Systems and Networks in today's scenario.

Outcome of the Course:

This skill-oriented course provides candidates with information required for configure and administration of OS Server, Networks, Database, Wireless & its Security and equip them with the skills required to protect & recover the computer systems & networks from various security threats in dedicated or virtual environment.

Full Flow of Course



Course Structure: This course contains total six modules. After completing the first five modules, the students have to do a 30 Hrs. Project using any of the topics studied to get certified.

IS281	Module Name	Duration (in Weeks)
Module 1	Fundamentals of Network and Operating Systems	2 Weeks
Module 2	Fundamentals of Cyber Security	2 Weeks
Module 3	Cryptography and Ethical Hacking	2 Weeks
Module 4	Network and Infrastructure Security	2 Weeks
Module 5	Vulnerability Assessment and Penetration Testing	2 Week
Module 6	Cyber Forensics	2 Week
Module 7	Project Work	1 Week
Module 8	Employability Skills	1 Week
Total Duration		14 Weeks

Course Fees

Registration Fee- Rs.1000/-

Course fee is Rs. 23,800/- {Can be paid as a single instalment of Rs.24, 200/- (Including Reg. fee) or in 2 instalments as given below}

Registration Fee	Rs. 1000/-	Nil/-	
Installment No	General Candidates (Course Fee.)	SC-ST Candidates (Course Fee.)	Last Date
1	11,900.00	Nil/-	10-09-2025
2	11,900.00	Nil/-	10-10-2025
Total	24,800.00	Nil/-	

Registration Fee- Refund Policy:

(Non-Refundable if candidate is selected for admission but did not join and if a candidate has applied but not eligible.)

However, the registration fee shall be refunded on few special cases as given below:

- ✓ Candidates are eligible but not selected for admission.
- ✓ Course postponed and new date is not convenient for the student.
- ✓ Course cancelled.

Eligibility

- ✓ Diploma – Pursuing/completed, Grade 10th completed, Grade 8th completed.

How to Apply?

Candidates can apply online in our website <https://nva.nielit.gov.in> Payment towards non-refundable registration fee can be paid through any of the following modes:

- ✓ Online transaction: Account No: 31185720641 Branch: Kottur (Chennai), IFS Code: SBIN0001669.
- ✓ Payment Gateway/GPAY/any UPI

Note: The Institute will not be responsible for any mistakes done by either the bank concerned or by the depositor while remitting the amount into our account

Last date of Registration: 08th September 2025

Registration Procedure

All interested candidates are required to fill the Registration form online with registration fee **On or before 08th September 2025** with all the necessary information.

Selection Criteria of candidates

Selection of candidates will be based on their marks in the qualifying examination subject to eligibility.

- ✓ The first list of Provisionally Selected Candidates will be published on NIELIT Chennai website (www.nielit.gov.in/chennai) on **09th Sep 2025** by **5:00 PM**. In case of vacancy, an additional selection list will be prepared and the selection will be intimated by email only.
- ✓ Provisionally selected candidate has to upload following document on registration portal for online verification :
 - Original Copies of Proof of Age, Qualifying Degree (Consolidated Mark sheet & Degree Certificate/Course Completion Certificate), 10th and 12th mark sheets.
 - Self-attested copy of community certificate.
 - One passport size photograph.
 - **AADHAR Identity proof.**
- ✓ After document verification **selected candidates have to pay first instalment of Rs. 11,900/- (for General/OBC/EWS/SC/ST)** on or before **10th Sep 2025** by payment mode mentioned above.
- ✓ Selected candidates are requested to upload the proof of remittance of fee on registration portal and also send the proof of remittance of fee as email to **pdheen@nielit.gov.in / trng.chennai@nielit.gov.in.**

Admission: All provisionally selected candidates whose documents are verified and paid the fees (full or first instalment) and verified by accounts section of NIELIT Chennai will get a welcome message in his login id provided during registration. The credential and url for online portal will be shared through WhatsApp or email.

Note: Provisionally Selected candidates have to visit NIELIT Chennai for Biometric and Certificate Verification including for Final Examination. If Verification is incomplete their candidature will be cancelled without any intimation.

Discontinuing the course

- ✓ No fees (including the security deposit) under any circumstances, shall be refunded in the event of a student who have completed the process of admission or discontinuing the course in between. No certificate shall be issued for the classes attended.
- ✓ If candidates are not uploading consecutive 3 assignments within assigned time, then their candidature will be cancelled without any notice and all fees paid will be forfeited.
- ✓ If candidates are not appearing for any internal examinations/practical their candidature will be cancelled without any notice and all fees paid will be forfeited.

Course Timings: This program is a practical oriented one and hence there shall be more lab than theory classes. The classes and labs are online cloud based from 9.30 am to 5:30 pm and Monday to Friday.

Address:

National of Electronics and Information Technology
ISTE Complex, No. 25, Gandhi Mandapam Road, Chennai – 600025
E-mail: trng.chennai@nielit.gov.in/Phone: 044-24421445
Contact Person: Dr. Munivel . E, Mobile: 9443349257

Course enquiries

Students can enquire about the various courses either on telephone or by personal contact between 9.15 A.M. to 5.15 P.M. (Lunch time 1.00 pm to 1.30 pm) Monday to Friday.

Placement:

Students who have completed the course successfully and qualified, Placement guidance and career counselling will be given to crack their interviews.

Important Dates

- ☐ **Last Date of Registration: 08th September 2025**
- ☐ **Display of Provisional Selection List: 09th September 2025**
- ☐ **Payment of first instalment fee: 09th September 2025 to 10th September 2025**
- ☐ **Course Start Date: 10th September 2025**
- ☐ **Payment of second instalment fee: within 10th October 2025**

Examination & Certification

- ✓ Certification Body: Examination Section, NIELIT HQ
- ✓ Certificates will be issued after successful completion of all the modules including project. For getting Certified Cloud Computing Engineer certificate a candidate has to pass each module individually with minimum required marks

Examination Pattern:

Examination scheme is as follows:

Theory (Each Question will carry 1 mark) Duration (in Min): 90		Practical			Internal Assessment (Marks)	Major Project/ Dissertation		Total
Papers	No. of Question Per Paper	Papers	Duration (in Min)	Marks Per Paper		No. of Projects	Marks	
2	100	1	90	90	30	1	30	350

Theory Papers

- Theory 1 – Cyber Security Associate Paper -1
- Theory 2 – Cyber Security Associate Paper -2

Practical Papers

- Practical 1 - Cyber Security Associate

Means of Assessment					
	Theory	Practical	Internal Assessment	Major Project/Dissertation	
No of Papers	02	01	01	01	
Marks Each Paper	100	90	30	Presentation	Projects
				15	15
Total Marks	200	90	30	30	
				350	

Grading Scheme

Following Grading Scheme (on the basis of total marks) will be followed:

Grade	S	A	B	C	D	Fail
Marks Range (in %)	85 to 100	75 to 84	65 to 74	55 to 64	50 to 54	Below 50

Final Grading as per above grading scheme will be given on the basis of total marks obtained in all modules.

Faculty

Mr. Yatish Kumar. P	Mr. Sakthi Guru. N.S	Mr. Mohamed Affan. R
BE (CSE)	B.Tech (CSE)	BE (CSE)
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Detailed Syllabus

Module1: Fundamentals of Network and Operating Systems

- Computer Networking
- OSI Reference Model & TCP/IP Protocol Suite
- TCP/IP command line Troubleshooting utilities
- Overview and Exploring Windows OS
- Overview and Exploring Linux OS

Module 2: Fundamentals of Cyber Security

- Basics of Information Security
- Ethical Hacking and Basic Cryptography
- Introduction to OS security
- Security Tools for Windows Operating System
- Introduction to Wireless Network and Wireless Security
- Introduction to the Android and iPhone iOS Operating systems
- Android and iPhone iOS Operating systems
- Web Application & Terminologies
- Web Application Protocols & Browser Security
- Introduction to Social Media & its security
- Digital Payment System and Security
- Various kinds of fraud in banking
- Introduction to Cyber Crime
- Overview on Cyber Law & Cyber Cell
- Cyber Fraud Helpline

Module 3: Cryptography and Ethical Hacking

- Introduction to Cryptography
- Public Key Infrastructure (PKI)
- Scanning and Information Gathering
- Open-source intelligence (OSINT)
- Man in the Middle (MITM) Attacks
- Password Cracking and Countermeasures
- Denial of Service and Countermeasures
- Steganography
- E-Mail Security

Module 4: Network and Infrastructure Security

- Introduction to Network Security and L2/L3 Attacks
- Network Security Control
- Intrusion Detection System(IDS)
- Understanding SOC

Module 5: Vulnerability Assessment and Penetration Testing

- Vulnerability Assessment and Exploring OWASP
- Introduction to Penetration Testing and VAPT Tools

Module 6: Cyber Forensics

- Introduction to Digital Forensics
- Digital Forensics & Acquisition
- File & Disk Analysis
- Memory Forensics
- System & Network Forensics
- Logs & Reporting

Module 7: Project Work

- Problem Definition and Architectural Development
- Requirement Analysis
- Security Server Selection
- Configuring Server
- Final Design & Deployment Phase
- Testing Phase

Module 8: Employability Skills

- Career Development & Goal Setting
- Essential Digital Skills
- Getting ready for Apprenticeship & Jobs